Fiscal Year:	FY 2021	**Task Last Updated:**	FY 03/31/2021
PI Name:	Pluth, Janice Ph.D.		
Project Title:	Space Adaptation Effects on Immune System Impacts Reproductive Function and Mammary Gland Development Across Generations		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology (2) Developmental Biology (3) Immunology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	janice.pluth@unlv.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	702-895-3236
Organization Name:	University of Nevada, Las Vegas		
PI Address 1:	Department of Health Physics and Diagnostic Sciences		
PI Address 2:	4505 S. Maryland Pkwy, Mail Code: 3037		
PI Web Page:			
City:	Las Vegas	**State:**	NV
Zip Code:	89154-9900	**Congressional District:**	1
Comments:	NOTE: Formerly at Lawrence Berkeley National Laboratory until fall 2017.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	10/04/2017	**End Date:**	12/31/2020
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
Contact Email:	Yuri.V.Griko@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 12/31/2020 per NSSC information (Ed., 9/10/2020) NOTE: End date changed to 9/01/2020 per NSSC information (Ed., 4/27/2020) NOTE: End date changed to 3/31/2020 per NSSC information (Ed., 6/4/19) NOTE: End date changed to 7/30/2019 per NSSC information (Ed., 7/13/18) NOTE: New grant number added and end date changed to 7/30/2018 per F. Hernandez/ARC (Ed., 4/13/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC18K0373 ; 80NSSC18K0019		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

[ED. NOTE 12/12/2017: Continuation of "Space Adaptation Effects on Immune System Impacts Reproductive Function and Mammary Gland Development Across Generations," per PI move to University of Nevada, Las Vegas from Lawrence Berkeley Laboratory.]
Space travel results in adaptation to unique conditions that include exposure to low doses of cosmic radiation and microgravity, and heightened stress. Adapting to these physiological and psychological stressors can alter hormone and immune functions, disrupting normal development of hormone-sensitive organs and negatively impacting reproductive health. Studies investigating mammary gland development in young adult mice will help define how adaptations to space travel can influence these fundamental processes. The ductal network of the mammary gland contains a number of different cell types including epithelial cells, adipocytes, immune cells, vascular endothelial cells, and fibroblasts. Signaling between these cell types is highly sensitive to environmental perturbations. In addition, both hormone and immune responses have been shown to play a critical role in ductal network branching of the developing mammary gland. A combination of novel assays will be used to understand how space exposures impact immune function adversely affecting mammary gland development, stem cell populations, and the reproductive ability of mice. We will test whether space adaptation-induced epigenetic changes in the oocytes of mice correlate with reproductive ability. We hypothesize that the space environment weakens the immune system, and alters hormone balance resulting in alterations in number and function of mammary stem cells and development of the mammary gland. Additionally, we hypothesize that epigenetic changes as a result of space exposure will affect reproductive capability as compared to mice on Earth. We further propose that the sensitivity to these effects is dependent on the genetic background of exposed mice. These studies will aid in defining how alterations in immune function from adaptation to space travel and subsequent re-adaptation on Earth, result in defects in organ development, reproductive ability, and epigenetic changes affecting the fitness of subsequent generations.

July 2016: Definition stage, ground-based experiments to prepare for in-flight space studies. Hypothesis for ground-based study: We will test the hypothesis that the genetically driven sensitivity of the immune system affects mammary gland development post fractionated Low LET (low energy transfer) radiation exposures. In space flight, mice are subject to chronic low-dose radiation exposures. This low dose of radiation exposure may have a significant impact on endocrine function and immune response and influence organ development. Prior to in-flight experiments it would be useful to perform ground-based studies using fractionated X-ray exposures to streamline procedures and to obtain information regarding how genetic background and associated immune system response impacts mammary gland development following radiation exposures. To answer this, we propose to test our hypothesis with the following specific aims in these modified ground-based studies:

AIM 1: To determine how low-dose fractionated radiation affects immune function in young adult mice and its consequences on mammary organ development.

AIM 1.1: To examine acute effects of immune response to low-dose fractionated radiation on mammary gland development.

AIM 1.2: To investigate how genetic background modulates the biological effects of fractionated radiation.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

FINAL REPORTING APRIL 2021
Grant Objectives: The intent of the original inflight proposal was to define the effects of space exposure on organ development. However, it was noted following the start of the project that in-flight studies would not be possible due to the International Space Station (ISS) mouse cages not able to accommodate the small, early aged mice needed for this study. Thus, we modified the proposal to a fully ground-based study which would still provide novel and useful, publishable information. To better understand how dose rate effect which are important aspects for NASA, we chose to compare a low acute (0.5 Gy) vs low fractionated (0.1 Gy x 5) exposure. The mammary gland is an organ that develops after birth and thus is a good model to study for this work. We assayed dose rate effects on immune response and mammary gland development using two genetically diverse strains of mice. For ease, X-ray was used as the radiation source as we were concentrating on dose rate effects rather than radiation quality effects. One strain of mouse chosen (BALB/c) has a greater sensitivity to radiation than the other strain (C57BL/6) to roughly mimic differences in genetic sensitivity in the population. In space, mice and astronauts would be subject to a chronic low dose exposure to high LET (linear energy transfer) radiation. This low dose of radiation could have a significant impact on endocrine function and immune response and influence a developing organ, especially for the mammary gland, which is dependent upon immune cell populations for proper development. Low dose fractionated X-ray exposures here on Earth could provide some insight into how the protracted low doses received in space might affect mammary organ development. In total, these ground-based studies will allow us to obtain information regarding how genetic background and radiation effects on the immune system responses impact mammary gland development following fractionated or acute low LET radiation exposures.

Specific Aims of the ground based study were the following: AIM 1: To determine how low-dose fractionated radiation affects immune function in young adult mice and its consequences on mammary organ development. AIM 1.1: To examine acute effects of immune response to low-dose fractionated radiation on mammary gland development. AIM 1.2: To investigate how genetic background modulates the biological effects of fractionated radiation.

Overall progress / Accomplishments / Problems

In total from all three cohorts of mice treated, there were 108 mice analyzed (54 of each strain) for the various endpoints. The first cohort of mice consisted of a total of thirty-six mice, 12 sham, 12 fractionated, and 12 acutely exposed mice (6 of each strain). The second cohort of mice consisted of a total of 24 mice, 8 mice within each treatment group (4 for each strain). The third cohort was for the lipopolysaccharide (LPS) studies and consisted of 8 mice for each treatment group (half plus and minus LPS), for a total of 48 mice (24 of each strain). For all cohorts of mice, two weeks following radiation, or mock irradiation exposure mice were sacrificed (at 7 weeks of age) and mammary glands removed for various endpoints. As immune cells are key for proper gland development the study also defined differences in immune populations both in the blood and within the mammary gland, as well as systemic cytokine changes. To determine if the function of the immune cell population were changed by the regimen of radiation exposure, mice were challenged with LPS and immune populations and cytokines compared between groups. In terms

of the overall health of the animals, irradiated and mock-irradiated mice did not show any gross differences in appearance or functioning, nor would this have been expected given the very low doses of Xray used. However, weight gain, which is an indicator of overall health, was monitored throughout the experiment and revealed an interesting relationship between radiation regimen of exposure, weight gain post radiation exposure, and the strain of mouse. In BALB/c mice, mock-irradiated mice showed the greatest body weight gain, followed by the fractionated radiation exposed mice. BALB/c mice exposed to an acute dose showed the lowest weight gain, significantly different from either of the other groups ($p<0.001$). In contrast, C57BL/6 mice showed the opposite response, with the sham irradiated group showing the lowest weight gain, the fractionated group showing a significantly higher level, and the acutely exposed group showing a significantly greater weight gain than the other groups ($p<0.001$). Consistent with our findings in C57BL/6 mice, other work has shown significant radiation-induced increases in the body weight of female C57BL/6 mice exposed to split-doses (Babbitt et al., 2001), noting increased adiposity (Nakamura et al., 2010) in mice exposed to fractionated radiation. These types of studies have not been done with the BALB/c strain, thus, the impact on this strain in response to radiation is a new finding. Our results revealing radiation-induced differences in weight gain based on the regimen of exposure and variant effects based on genetics, fits in part with prior studies; however, the mechanism underlying the strain-dependent patterns is unclear and would require further studies. Complete blood cell counts (CBC) were determined for each mouse to define if the radiation-regimen of exposure persistently altered immune populations. We observed the BALB/c strain to show a larger number of immune cell populations with significant differences between the radiation exposed groups and controls as compared to the C57BL/6 strain. BALB/c mice exposed to a fractionated dose showed significantly higher percentages of eosinophils as compared to both sham, and lower levels of platelets as compared to acutely exposed. Following acute exposures BALB/c mice showed significantly lower mean platelet volume (MCV) than acute exposed mice as well as higher mean corpuscular volume as compared to sham or fractionated. Similar trends were often observed in the C57Bl/6 mice but they did not reach significance with the exception of the mean platelet volume, which in acutely exposure mice showed significantly lower levels as compared to sham and the fractionated groups. Additionally, in C57BL/6 mice there was a significant difference between sham and acute, with acute showing lower level of platelets as compared to sham.

Cytokines are important in restoring homeostasis following exposures. To define if different exposures caused lasting changes in cytokines weeks post exposure, a multiplexed antibody assay was used to define the plasma level of 32 cytokines. Only C57BL/6 mice showed significant persistent changes in cytokines two weeks post exposure (in 5 of the 32 cytokines). For example, GM-CSF showed a significant decrease in acutely exposed mice as compared to sham irradiated. IL-1b levels in C57BL/6 mice exposed to the fractionated regimen were significantly elevated over acute, with the acutely irradiated animals also elevated over control, but not to a significant degree. Significant elevations of IL-17 were observed in C57BL/6 mice exposed to the fractionated regimen as compared to sham. Plasma levels of TNFa were increased in C57BL/6 mice exposed to the fractionated low-dose radiation as compared to acutely exposed C57BL/6 mice. C57BL/6 mice exposed to fractionated doses also showed lower levels of IL-12 (p40) as compared to both sham and acute. Although low-dose radiation did not alter the plasma levels of any of the measured cytokines in the BALB/c strain, IL-17 and TNFa showed a trend similar to C57BL/6 mice. Overall, the majority of the significantly altered cytokines were observed in the fractionated exposed C57BL/6 mice, indicating that fractionated exposures, even at doses as low as 10 cGy/ day, produce lasting effects (weeks post exposure).

Environmental exposures and alterations in immune cell populations can impact ductal branching and sub-organelle development within the mammary gland. To determine whether the radiation exposure regimen impacted the development of structures within the mammary gland, laser scanning tissue autofluorescence and fluorescence imaging (LSTAFI) was used to look for small defects as well as differences in ductal growth and branching. Overall ductal length measurements were not significant between groups, however mammary ductal branching as well as the size and number of terminal end buds (TEB) in the mammary gland differed based on dose rate and genetics. The degree of branching was quantified throughout the mammary gland to define if branching were altered with low doses of radiation and showed differences dependent on radiation regimen and genetic background. Significant differences were observed for both strains exposed to the fractionated regimen, although the strains showed an opposite pattern. BALB/c mice exposed to a fractionated regimen showed a significant increase in branching in the mid-region of the gland, whereas fractioned C57BL/6 mice showed significantly less branching than sham and or acutely exposed mice. We conclude that low-doses of radiation do not affect ductal length, but do significantly affect branching morphogenesis when radiation exposures occur during puberty.

As hormone levels are important in branching morphogenesis (Satoh et al., 2007; Sternlicht et al., 2006), we investigated whether there were significant differences in estradiol or progesterone levels between the strains or the groups; however, there were no significant differences in either strain. It’s possible, however, that other growth factors, perhaps fibroblast growth factor (FGF), a growth factor known to be important in branching, may be altered uniquely with exposure. Further studies would be needed to dissect out the role of other growth factors on the branching differences we observed based on genetics and radiation dose rate. To define how terminal endbuds (TEBs), important organelles containing stem cells that initiate the progression of the duct across the fat pad, were specifically impacted by exposure individual TEBs were analyzed from three representative images per mouse to detect morphological defects. Two different defects were specifically focused on, thinning of the ducts and breakage of the ducts near the TEB region. Results revealed greater numbers of thinning and broken ducts in the neck region near the TEB in the radiation exposed mice of both strains when compared to sham. Significantly greater percentages of TEB with thinning ducts were observed in C57BL/6 mice acutely exposed to radiation, although the fractionated group also had similar percentages. In addition, greater numbers of ductal breaks were observed in both radiation groups as compared to sham, although this was not significant. Exposed BALB/c mice also showed a non-significant trend of higher numbers of thinning ducts. Greater percentages of mice from the exposed groups revealed one or both of these defects. Therefore, low doses of radiation result in an increased level of defects near the TEB, but the extent is dependent on genetics. Given TEB are present during specific times during development and should grow, mature, and disappear at a specific rate, the size of the TEB provides an indicator of the developmental process. Thus, the diameters of all end buds within the distal 2 mm region of the furthest endbud were measured and endbuds (EB) >100 um (defined based on size as TEBs), were quantified and binned by size. Significantly higher percentages of TEBs were observed in BALB/c mice exposed to fractionated radiation ($p=0.006$) as compared to the sham irradiated group. Slightly, although not significantly, higher numbers of TEBs were also observed in the acutely exposed BALB/c mice as compared to the control. In contrast, no significant differences between groups were noted in C57BL/6 mice. When TEBs were binned by size, there was a larger percentage of TEBs of fractionated BALB/c mice showing greater diameters as compared to acute or sham exposed, although numbers did not reach significance. In comparison, C57BL/6 mice didn’t show larger sized TEBs in any group. The total area of the three largest TEBs for each strain was defined for each group and strain using IMARIS software. This revealed a trend for both strains whereby larger volumes were observed in the fractionated group, with a

Task Progress:

more dramatic effect in the BALB/c strain, although differences were not significant. Results reveal an increased size of TEB, specifically for BALB/c mice exposed to fractionated radiation, again pointing to the importance of the strain and radiation regimen of exposure in the detrimental effects observed. TEBs are typically more prevalent during puberty, and once they reach the outer edge of the fat pad they diminish in size and leave a blunt ended duct termini (Paine and Lewis, 2017). The fact that fractionated radiation results in a greater number of TEBs as compared to acute exposure, may suggest development has been stalled with fractionation, although total ductal length was not significantly different between strains and/or treatment groups. The reason for the greater number of TEBs is uncertain, but it could be due to a reprogramming of stem cells within the TEB, a possibility we would like to investigate further. Additionally, it may be due to changes in mammary gland immune cells that aid in TEB maturation. Immune cell populations infiltrating into the mammary gland are known to affect the number and size of TEBs6. Given BALB/c mice exposed to fractionated radiation showed significantly higher percentages of TEB in the distal region of the mammary gland, BALB/c mammary gland sections exposed to each radiation regimen were stained for immune T cell populations. We noted a significant difference in the number of CD4+ and CD8+ populations specifically in mice exposed to fractionated radiation as compared to acute or sham irradiated. Few CD4+ cells are typically observed in the normal mammary gland (Plaks et al., 2015), and this is reflected in BALB/c sham mice having on average only 0.62 CD4+ cells per image. In contrast, the BALB/c fractionated group had over twice the number of CD4+ cells. This was a significant difference observed only in fractioned mice that had shown greater numbers and larger sized TEBs.

Previous work has revealed CD4+ cells to provide signals that negatively regulate mammary organogenesis, which may fit with our finding of larger TEBs, whose inner stem cells did not correctly differentiate, yet proliferated. Although we have not exhaustively interrogated all aspects that may differ between the strains and be affected by fractionated exposure, this difference in the CD4+ T cells may provide a clue as to the higher number and larger size of TEBs within the gland of the BALB/c mice exposed to fractionated radiation. Unfortunately, due to COVID we have had some delays in our research, and thus are currently finalizing the last portion of our work testing whether the immune function of mice is compromised dependent on previous radiation exposure. We have used Lipopolysaccharide (LPS), a major microbial mediator for tissue injury resulting from a Gram-negative bacterial infection, to provide the challenge. Animals were exposed to LPS prior to sacrifice, 2 weeks post radiation, or mock irradiation. To verify the effect of LPS on the immune system we also tested plasma of mice at the time of sacrifice for serum amyloid A (SAA), which is robustly expressed with LPS treatment. All LPS treated mice showed significantly higher levels of SAA as compared to non-LPS treated mice. From the plasma at the time of sacrifice cytokines and CBC were also defined to determine if the immune function differed in response to LPS based on prior radiation exposure, weeks before. Given we did find evidence of radiation dose rate-dependent changes in the immune populations, we may expect some functional changes as well. Currently we are finishing the analysis of data from both strains of mice exposed to each radiation regimen plus and minus LPS treatment to define differences. Preliminary data suggests a functional defect, particularly in BALB/c mice, with a number of cytokines being aberrantly expressed as compared to Sham and Acute. In total the data thus far reveal low doses of radiation can induce long-term changes in the ability of cytokines to correctly respond to outside stimuli, and can be modified by dose rate and the underlying genetics. Similarly, CBC analysis shows changes in the proportions of various immune cells which are also dose rate and strain specific following LPS challenge. In total, the results of this study reveal even very low doses of X-ray, provided during a key window of development, can have lasting effects on the mammary gland. In addition, work reveals fractioned exposure is more damaging than an acute exposure in certain genetic backgrounds. Lastly, underlying differences observed may be attributed to differing levels of CD4+ T cells, which are higher within the mammary gland of BALB/c mice following fractionated radiation.

However, there are some limitations to these studies, including the fact that the percentage of other immune cell populations within the mammary gland, including both eosinophils and macrophages, immune cells important in mammary development, were not defined. Changes in the proportion of stem cells should also be investigated in further work to define if this plays a role in the differences observed with fractionated exposure, specifically in BALB/c mice. This work has clinical applications in considering radiation during puberty. Our work suggests that multiple closely-spaced CT scans may be more damaging than a single, same total dose exposure. Further studies would be needed to investigate the threshold dose eliciting this effect and whether number of fractions and timing between doses are key in the detrimental results. Of interest to NASA, these studies have provided crucial information regarding how genetics in combination with radiation may impact the immune system and disturb the growth and development of an organ. The low doses of acute and fractionated X-ray exposures would likely produce less of an effect as compared to high LET radiation in space, due to the lower level of biological effectiveness for low LET radiation. Thus, these data would predict that low levels of radiation in space, especially chronic radiation exposure, more similar to the fractionated radiation used in this study, may be more damaging to a developing organ. However, the degree of effects would likely be modulated depending on one’s genetics, as shown with the more severe effects in BALB/c mice as compared to C57BL/6 mice. Further studies comparing chronic vs acute high LET exposures would be needed to more accurately predict effects in space, but this work should provide some level of caution for exposures during this critical window of mammary development if people were ever to colonize planets in the future.

References

Babbitt, J. T. et al. Increased body weight in C57BL/6 female mice after exposure to ionizing radiation or 60Hz magnetic fields. International journal of radiation biology 77, 875-882, doi:10.1080/09553000110055790 (2001).

Nakamura, S. et al. Adiposity in female B6C3F1 mice continuously irradiated with lowdose-rate gamma rays. Radiation research 173, 333-341, doi:10.1667/RR1962.1 (2010).

Satoh, K. et al. Progesterone enhances branching morphogenesis in the mouse mammary gland by increased expression of Msx2. Oncogene 26, 7526-7534, doi:10.1038/sj.onc.1210555 (2007).

Sternlicht, M. D., Kouros-Mehr, H., Lu, P. & Werb, Z. Hormonal and local control of mammary branching morphogenesis. Differentiation; research in biological diversity 74, 365-381, doi:10.1111/j.1432-0436.2006.00105.x (2006).

Paine, I. S. & Lewis, M. T. The Terminal End Bud: the Little Engine that Could. Journal of mammary gland biology and neoplasia 22, 93-108, doi:10.1007/s10911-017-9372-0 (2017).

Plaks, V. et al. Adaptive Immune Regulation of Mammary Postnatal Organogenesis. Developmental cell 34, 493-504, doi:10.1016/j.devcel.2015.07.015 (2015).

ANNUAL REPORTING JULY 2020

Period of progress: 6/4/2019-07/30/2020

This ground-based study was initiated to provide relevant data for subsequent inflight experiments to follow. The intent of the original inflight proposal was to define the effects of space exposures on organ development. However, recently it was noted that in flight studies would not be possible due to the cages not being able to accommodate the small, early aged mice. Thus, we have modified the proposal to a fully ground-based study that will allow an important and novel published piece of work to result from these studies. In these studies, we have compared the effect of fractionated radiation vs acute exposure on immune response, stem cell populations, and mammary gland development using two genetically diverse strains of mice. One strain of mouse (BALB/c) has a greater sensitivity to radiation than the other strain (C57BL/6). The comparison of these two strains will allow us to define whether there is a genetic susceptibility to radiation effects. In space flight, mice will be subject to a chronic low dose exposure to radiation. This low dose of radiation may have a significant impact on endocrine function and immune response and influence organ development. Low dose fractionated exposures here on Earth could provide some insights as to how the protracted low doses received in space might affect organ development. Thus, these ground-based studies will allow us to obtain information regarding how genetic background and associated immune system response impact mammary gland development following fractionated or acute low LET radiation exposures.

Brief summary of past year's research accomplishments: Previously we noted with our first two cohorts, changes in weight dependent upon genetic background and radiation exposure. We also observed changes in immune cell populations and cytokine expression weeks post exposure that were also based on genetics and the radiation regimen of exposure. Defects within the mammary glands were also noted, particularly in the terminal end buds which were significantly larger in the radiation sensitive BALBc mouse following only fractionated radiation. We finished irradiation and collection of mammary glands for our last cohort of animals in 2019. Two strains of mice, one more genetically susceptible to radiation effects and the other more resistant, were exposed to different regimens of X-ray exposure during week 4-5 after birth.

In this past year in this final cohort we injected the animals with lipopolysaccharide (LPS) within the last 3 h before sacrificing (at 7 weeks). Lipopolysaccharide (LPS) is a constituent of the outer cell membrane of Gram-negative bacteria and is a highly potent proinflammatory substance. This will help to establish if the low dose irradiation regimens they had received earlier in life have compromised the function of their immune system. We also would with this final cohort like to perform stem cell studies to define if the larger terminal end buds (TEB) we note in the BALBc mice following the fractionated exposure may be due to defective stem cell function. We are now in the process of analyzing these specimens. We will also perform analysis of LSTAFI for defects in the glands as well as immune cell analysis to compare to previous findings.

The analysis of the LPS challenge studies will add to what we had already defined and help to establish that exposure alters immune system function and is dependent upon the radiation regimen of exposure and genetics of the mice.

Given issues with COVID-19 and the shutdown of the lab we have not been able to process samples on schedule but hope to slowly be able to do so in the upcoming months. This shutdown will now require until Dec. 2020 to complete the work we had planned to do from March-July of this year. This final portion of the work will provide crucial information regarding how genetics in combination with radiation may impact the immune system function and disturb the growth and development of an organ. The very low doses of fractionated radiation will provide data mimicking the low chronic doses in space and provide information regarding what may be expected with space travel.

Bibliography Type:	Description: (Last Updated: 11/01/2023)
Articles in Peer-reviewed Journals	Wiedmeyer B, To J, Sridharan DM, Chien LC, Snijders AM, Mori H, Pluth JM. "Dose fractionation during puberty is more detrimental to mammary gland development than an equivalent acute dose of radiation exposure." Int J Radiat Oncol Biol Phys. 2021 Apr 1;109(5):1521-32. Epub 2020 Nov 21. https://doi.org/10.1016/j.ijrobp.2020.11.039 ; PMID: 33232771 , Apr-2021